

VISHNU TIWARI

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EDUCATION

JSS Academy of Technical Education, Noida — B.Tech in Information Technology | Expected Aug 2028

GPA: 8.5 / 10.0 | Relevant Coursework: Data Structures & Algorithms, OOP (Java), Statistical Analysis

TECHNICAL PROJECTS

Neural Machine Translation — Seq2Seq with Bahdanau Attention

Tech Stack: Python, PyTorch, LSTM/GRU, Bahdanau Attention, Jupyter Notebook

GitHub: github.com/visshhnnuuuuu/seq2seq-from-scratch

- Designed and implemented an end-to-end NMT system from scratch using PyTorch, incorporating a Seq2Seq Encoder-Decoder architecture with Bahdanau Attention to enable dynamic context alignment.
- Built custom tokenization, vocabulary construction, and padding pipelines; applied Teacher Forcing during training to accelerate convergence on English-to-Hindi translation.
- Attention layer computes alignment scores and context vectors per decoding step, significantly improving translation quality on long sequences over vanilla Seq2Seq baselines.

Deep Reinforcement Learning — Frozen Lake Agent

Tech Stack: Python, PyTorch, OpenAI Gymnasium, Q-Learning, DQN

GitHub: github.com/visshhnnuuuuu/Frozen-lake-DRL

- Trained a Deep Q-Network (DQN) agent to navigate FrozenLake-v1, implementing experience replay and epsilon-greedy exploration to balance exploitation vs. exploration.
- One-hot encoded discrete state observations to enable stable neural network training across the 16-state stochastic grid environment.
- Tracked reward curves and episode lengths across runs to evaluate agent performance and tune hyperparameters (learning rate, discount factor, replay buffer size).

Credit Card Churn Prediction — ANN

Tech Stack: Python, TensorFlow/Keras, Pandas, Scikit-Learn, Matplotlib

GitHub: github.com/visshhnnuuuuu/Credit-Card-churn-prediction-through-ANN

- Built a Multi-Layer Perceptron (ANN) on an imbalanced 10,000-record dataset, applying SMOTE oversampling to address 16% minority class and improve recall on at-risk customers.
- Performed end-to-end feature engineering: encoding of high-cardinality categorical variables, feature scaling, and correlation-based selection to optimize model input quality.
- Prioritized recall as the primary business metric to maximize identification of churners and enable proactive customer retention strategies.

Nerd Prediction Web Application

Tech Stack: Python, Scikit-Learn, Pandas, Streamlit, Pickle

GitHub: github.com/visshhnnuuuuu/nerd-prediction-app

- Engineered an end-to-end ML pipeline to classify user categories from behavioral flow data; deployed Logistic Regression model via Streamlit Cloud with real-time low-latency inference.
- Applied Feature Scaling and Label Encoding during preprocessing, achieving high model compatibility with minimal overfitting through cross-validation.

Exploratory Data Analysis & Feature Engineering Suite

Tech Stack: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn

GitHub: github.com/visshnnuuuuu/EDA

- Black Friday Analysis: Cleaned high-cardinality retail data using Scikit-Learn imputers, applied feature scaling/encoding, and identified key purchase drivers through correlation heatmaps.
- Zomato Consumer Insights: Applied One-Hot Encoding and normalization to standardize diverse features and surfaced key behavioral drivers through multivariate visualizations.

TECHNICAL SKILLS

Languages: Python (Advanced), Java (Core)

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn

NLP / RL: Seq2Seq, Attention Mechanisms, Transformer Concepts, Q-Learning, DQN

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Tools & Platforms: Git, GitHub, Jupyter Notebook, Streamlit, Kaggle, VS Code

LEADERSHIP & ACTIVITIES

- Runner-up, Shiv Nadar University Sports Fest (Badminton) | 2024
- Member, PFC (Photography Club) — JSSATE
- Member, JSS Badminton Club